

MATH120-24S2 Discrete Mathematics

Tutorial 4

References

- Topic 5 and 6: Divisibility and Prime factorisation
- Sections 4.4, 4.10, 8.4 of Epp (5th Ed)
- Sections 4.3, 4.8, 8.4 of Epp (4th Ed)

Work on the following problems in groups during the tutorial in week 5.
Start with the marked questions to ensure good coverage of the material.

- ⇒ 1. [B] If $a, b, c \in \mathbb{Z}^+$ and $a|bc$, does it follow that $a|b$ or $a|c$? Prove or give a counterexample.

Solution:

If $a, b, c \in \mathbb{Z}^+$ and $a|bc$, it does not follow that $a|b$ or $a|c$. For example, $6|(2 \times 3)$ but $6 \nmid 2$ and $6 \nmid 3$.

2. [B] Factorise 2020 in prime numbers. Use trial division to show that all the factors you write down are in fact primes.

Solution:

$2020 = 2^2 \cdot 5 \cdot 101$. We know that 2 and 5 are prime. To check that 101 is prime, we note that we only need to check possible prime numbers up to $\sqrt{101}$. It is readily checked that $2 \nmid 101, 3 \nmid 101, 5 \nmid 101, 7 \nmid 101$.

3. [M] If $a, b \in \mathbb{Z}^+$ with $a = 630$, $\gcd(a, b) = 105$, and $\text{lcm}(a, b) = 242,550$; what is b ?

Solution:

From Theorem 5.11 (in our notes) we know that $ab = \text{lcm}(a, b) \cdot \gcd(a, b)$. Consequently,

$$b = \frac{\text{lcm}(a, b) \cdot \gcd(a, b)}{a} = \frac{242,500 \cdot 105}{630} = 40,425.$$

- ⇒ 4. [B] For each of the pairs $a, b \in \mathbb{Z}^+$,

(a) $a = 231, b = 1820$

(b) $a = 1369, b = 2597$

do the following:

(a) determine $\gcd(a, b)$,

(b) express $\gcd(a, b)$ as a linear combination of a, b ,

(c) find $\text{lcm}(a, b)$.

Solution:

(a) i. Applying the Euclidean algorithm, we have

$$1820 = 7(231) + 203 \quad (1)$$

$$231 = 1(203) + 28 \quad (2)$$

$$203 = 7(28) + 7 \quad (3)$$

$$28 = 7(4).$$

The last non-zero remainder is 7, so $\gcd(1820, 231) = 7$.

ii. Starting from (3) in part (i) and working backwards, we get

$$7 = 203 - 7(28) \quad \text{from (3)}$$

$$= 203 - 7(231 - 203) \quad \text{from (2)}$$

$$= (-7)(231) + 8(203)$$

$$= (-7)(231) + 8(1820 - 7(231)) \quad \text{from (1)}$$

$$= 8(1820) + (-63)(231).$$

iii. From Theorem 5.11 (in our notes), we know that $ab = \text{lcm}(a, b) \cdot \gcd(a, b)$. Consequently,

$$\text{lcm}(231, 1820) = (231)(1820)/7 = 60,060.$$

(b) i. Applying the Euclidean algorithm yields $\gcd(1369, 2597) = 1$.

- ii. Reversing the steps of the Euclidean algorithm in part (i), we obtain

$$1 = 2597(534) + 1369(-1013).$$

- iii. By Theorem 5.11 (in our notes),

$$\text{lcm}(1369, 2597) = (1369)(2597) = 3, 555, 293.$$

$\Rightarrow$ 5. [M] Prove the following statements.

- (a) For all $a, b \in \mathbb{Z}^+$, $a|b$ if and only if $\text{lcm}(a, b) = b$.
 (b) For $a, b, c \in \mathbb{Z}^+$, we have that $\text{gcd}(\text{gcd}(a, b), c) | \text{gcd}(a, \text{gcd}(b, c))$.

Solution:

- (a) ($\rightarrow$) Assume that $a|b$. Since it is also the case that $b|b$, we have that b is a common multiple of a and b . Now let c be the least common multiple of a and b , then $a|c$ and $b|c$. In particular, we find that $b \leq c$, and hence, since b is a common multiple and c is the least common multiple, we find that $c = b$.
 ($\leftarrow$) Assume that $\text{lcm}(a, b) = b$ for $a, b \in \mathbb{Z}^+$. By definition, $a|\text{lcm}(a, b)$ and hence since $\text{lcm}(a, b) = b$, $a|b$.
 (b) First recall that if $y|\text{gcd}(a, b)$ then $y|a$ and $y|b$, and moreover, that if $e|f$ and $e|g$ then $e|\text{gcd}(f, g)$. Let $x = \text{gcd}(\text{gcd}(a, b), c)$, then $x|\text{gcd}(a, b)$ and $x|c$. Since $x|\text{gcd}(a, b)$ we have that $x|a$ and $x|b$. So we find $x|a, x|b, x|c$. Since $x|b, x|c$, it follows that $x|\text{gcd}(b, c)$. And since $x|a$ and $x|\text{gcd}(b, c)$, it follows that $x|\text{gcd}(a, \text{gcd}(b, c))$. We conclude that indeed $\text{gcd}(\text{gcd}(a, b), c) | \text{gcd}(a, \text{gcd}(b, c))$.

6. [B] Given the two integers 7, 114, 800 and 7, 882, 875:

- (a) write down both integers as a product of primes,
 (b) find the number of positive divisors for each integer,
 (c) determine their greatest common divisor,
 (d) determine their least common multiple.

Solution:

- (a) $7114800 = 2^4 \cdot 3 \cdot 5^2 \cdot 7^2 \cdot 11^2$, $7882875 = 3^2 \cdot 5^3 \cdot 7^2 \cdot 11 \cdot 13$
 (b) $7114800 = 2^4 \cdot 3 \cdot 5^2 \cdot 7^2 \cdot 11^2$ has $(4+1)(1+1)(2+1)(2+1)(2+1) = 270$ positive divisors. $7882875 = 3^2 \cdot 5^3 \cdot 7^2 \cdot 11 \cdot 13$ has $(2+1)(3+1)(2+1)(1+1)(1+1) = 144$ positive divisors.
 (c) $\gcd(2^4 \cdot 3 \cdot 5^2 \cdot 7^2 \cdot 11^2, 3^2 \cdot 5^3 \cdot 7^2 \cdot 11 \cdot 13) = 3 \cdot 5^2 \cdot 7^2 \cdot 11 = 40425$
 (d) $\text{lcm}(2^4 \cdot 3 \cdot 5^2 \cdot 7^2 \cdot 11^2, 3^2 \cdot 5^3 \cdot 7^2 \cdot 11 \cdot 13) = 2^4 \cdot 3^2 \cdot 5^3 \cdot 7^2 \cdot 11^2 \cdot 13 = 1387386000$

$\Rightarrow$ 7. [M] Show the following.

- (a) If n is any positive integer whose last digit is a 5, then $5|n$.
 (b) An integer is divisible by 9 if and only if the sum of its digits is divisible by 9.

Solution:

- (a) **Proof:** Suppose that n is a positive integer whose last digit is a 5. Then

$$n = 10m + 5$$

for some integer m .

$$n = 10m + 5 = 5(2m + 1),$$

where $2m+1$ is an integer since m is an integer. Hence n is divisible by 5 (by definition of divisibility).

- (b) Let $n = a_t a_{t-1} a_{t-2} \dots a_1 a_0$ be an integer, where a_j denote the digits of n , then n can be written in decimal expansion as

$$\sum_{i=0}^t a_i 10^i.$$

Now for all i ,

$$10^i \equiv 1 \pmod{9},$$

which implies that

$$\sum_{i=0}^t a_i 10^i \equiv \sum_{i=0}^t a_i \pmod{9},$$

which is precisely the sum of the digits of n . Hence, n is divisible by 9 if and only if the sum of its digits is divisible by 9.

8. (a) [D] What is a Mersenne prime?
 (b) [C] Let $m \in \mathbb{Z}^+$, $m > 1$. Show that $2^{2m} - 1$ is not a prime.

Solution:

- (a) A Mersenne prime is a prime number of the form $2^n - 1$ for some integer n .
 (b) We see that $2^{2m} - 1 = (2^m + 1)(2^m - 1)$. Since $m > 1$, we have that both factors are larger than one, which implies that $2^{2m} - 1$ is composite.
9. [C] Let $n!$ denote the product of the integers 1 through n .
 (a) What is the remainder when $(1! + 2! + 3! + 4! + 5! + 6! + \dots)$ is divided by 9?
 (b) Show that there exists a prime p such that $n < p \leq n! + 1$.

Solution:

- (a) If $n \geq 6$, then $n!$ contains the factors 6 and 3, and hence, $n!$ is a multiple of 9. Thus, we only need to find $(1! + 2! + 3! + 4! + 5!) \bmod 9$. Now

$$\begin{aligned} 1! &\equiv 1 \pmod{9} \\ 2! &\equiv 2 \pmod{9} \\ 3! &\equiv 6 \pmod{9} \\ 4! &\equiv 24 \equiv 6 \pmod{9} \\ 5! &\equiv 120 \equiv 3 \pmod{9}. \end{aligned}$$

Hence, the remainder of $1! + 2! + 3! + 4! + 5! + \dots$ when dividing by 9 is $1 + 2 + 6 + 6 + 3 \bmod 9$, i.e. 0.

- (b) We will think about the possible prime factors of $n! + 1$. We first note that $n! + 1 = 1 + 1 \cdot 2 \cdot 3 \cdot \dots \cdot n$. So we can see that $n! + 1 \equiv 1 \pmod{2}$ and $n! + 1 \equiv 1 \pmod{3}$ and so on until $n! + 1 \equiv 1 \pmod{n}$. This means that $2 \nmid n! + 1$, $3 \nmid n! + 1, \dots$ and $n \nmid n! + 1$. Therefore, any prime p that divides $n! + 1$ must be greater than n and hence there exists (at least one) prime $n < p \leq n! + 1$.